Caleb Mowery

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Education

The Pennsylvania State University — University Park, PA
College of Engineering | B.S. in Computer Science — *Aug. 2023 – May 2027*
Minor in Entrepreneurship and Innovation — *GPA: 3.62/4.0*

Experience

Platform Engineering Intern — May 2026 – Present
IBM — *University Park, PA*

- Developed and deployed *Project Orchestrate*, an automation application integrating IBM's Salesforce implementation with watsonx Orchestrate to streamline Technical Sales Activity generation, saving each manager 1-2 hrs/week
- Led technical demonstrations of IBM's post-quantum cryptography solutions for 3 enterprise clients, developing technical documentation and presenting product capabilities, achieving a 100% adoption rate among clients engaged
- Co-architecte AI automation solutions integrating IBM watsonx with existing file-sharing, project management, and media production platforms, reducing client employee workloads by 1–5 hours per employee per week.
- **Skills:** Red Hat OpenShift, IBM watsonx Orchestrate, MCP Server Deployment, JavaScript, Sales

Director of Technology — Apr 2026 – Present
Penn State THON™ — *University Park, PA*

- Oversee all Technology usage for the largest student-run philanthropy in the world, which has raised over $273 million for Four Diamonds, a fund for cancer research and out-of-pocket cancer treatment expenses for Pennsylvania families
- Manage 26 student developers who are responsible for maintaining, optimizing, and enhancing thon.org, store.thon.org, our volunteer database, our THON event dashboard, and our AWS architecture
- Coordinate 16,500+ student volunteers, 25,000+ alumni supporters, and 600+ participating families, collaborating with a 19-member executive committee to support year-round fundraising and event operations
- **Skills:** AWS (EC2, ECS, RDS, CloudFront), Terraform, PowerShell, Docker, Python, Django, HTML/CSS

Research Assistant — Aug 2024 – Present
Pennsylvania State University — *University Park, PA and Kisumu, Kenya*

- Designed *Walimu Kwanza*, a web app used by 200+ teachers which allows Kenyan teachers to share lesson plans and classroom resources, as well as upload and answer questions to community forums, built using Adalo
- Researched teacher struggles in Kisumu, holding 2-4 stakeholder meetings a week focused on co-designing a digital solution to address issues with understaffing, low government funding, and a recent curriculum change in Kenya
- Co-authored a published research paper analyzing stakeholder interviews and Walimu Kwanza usage data to evaluate the effectiveness of educational technology in addressing challenges faced by teachers in low-resource Kenyan schools.
- **Skills:** Stakeholder Engagement, UI/UX Design, Adalo, Figma, Academic Writing

Projects

Multivac Hero | *Python, Go, React, JavaScript, OpenAI API, Three.js* — July 2026 – Present

- Built a full-stack application that converts uploaded music files into playable Guitar Hero guitar charts
- Integrated an audio-capable LLM with paired reference songs and MIDI charts to generate note timing and fret patterns
- Developed a Go API managing asynchronous Python jobs, file uploads, and token estimates before generation
- Implemented tempo-aware MIDI processing, selectable reference difficulty, and a chart view synchronized with audio

Recruitment Database | *Python, Google Apps Script* — May 2024 – Present

- Developed a scoring algorithm to evaluate student applicants to Alpha Kappa Psi, a business fraternity on campus
- Utilized Google Apps Script to generate scoring forms that show current members each applicant's photo and resume
- Form responses alongside interview scores are then ingested into a Python script that ranks applicants
- Slides of applicants are generated with GAS to show applicant rankings and resume highlights for deliberation

Honors and Interests

Honors: Eagle Scout Award, The President Walker Award
Interests: Breaking Bad, Guitar Hero, Leg Day, Noah Kahan, Outer Banks, Philosophy, Pearl Jam, Texas Hold 'Em